Supplementary material

Ian Driver*

Independent Researcher, Detroit, MI, USA
* Correspondence: driver.ian@gmail.com

Supplement to "A benchmark of cell-type annotation methods for single-cell data". Figures and tables referenced from the main text as Supplementary Figure S1–S11 and Supplementary Table S1–S3.

Supplementary figures

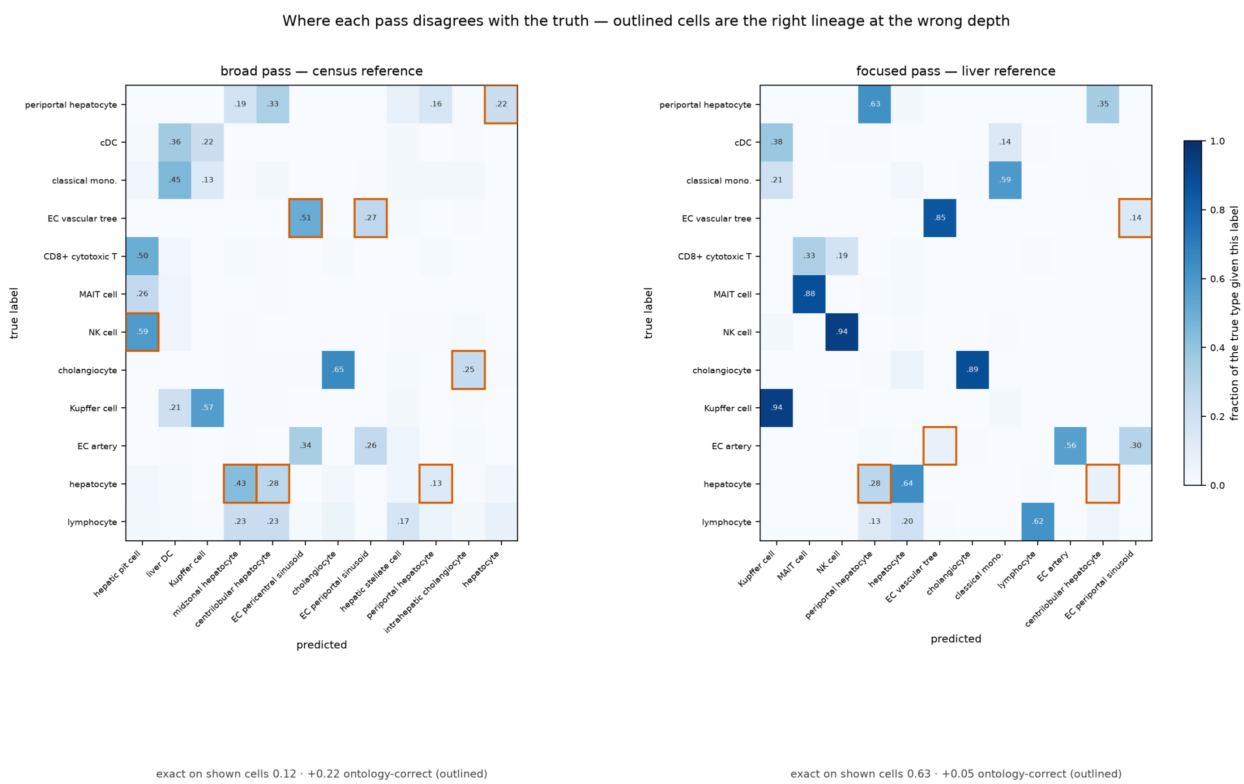

Figure S1. Confusion matrices for the broad and focused passes on the withheld cross-study liver query. Each cell is the fraction of a true type receiving that label; outlined cells are off-diagonal calls that are an ancestor or descendant of the truth in the Cell Ontology, i.e. error under exact match and credit under ontology-aware concordance. The scores below each panel are means over the twelve truth types drawn, not over the whole query, so they run lower than the query-wide figures quoted in the main text. The broad pass scores 0.12 exact and recovers +0.22 through the ontology — its mistakes are the right lineage at the wrong depth (hepatocyte → midzonal or centrilobular hepatocyte; NK cell → hepatic pit cell, the Cell Ontology term for a liver NK cell).

The focused pass scores 0.63 exact, +0.05: it is already right most of the time, so the ontology has little left to recover.

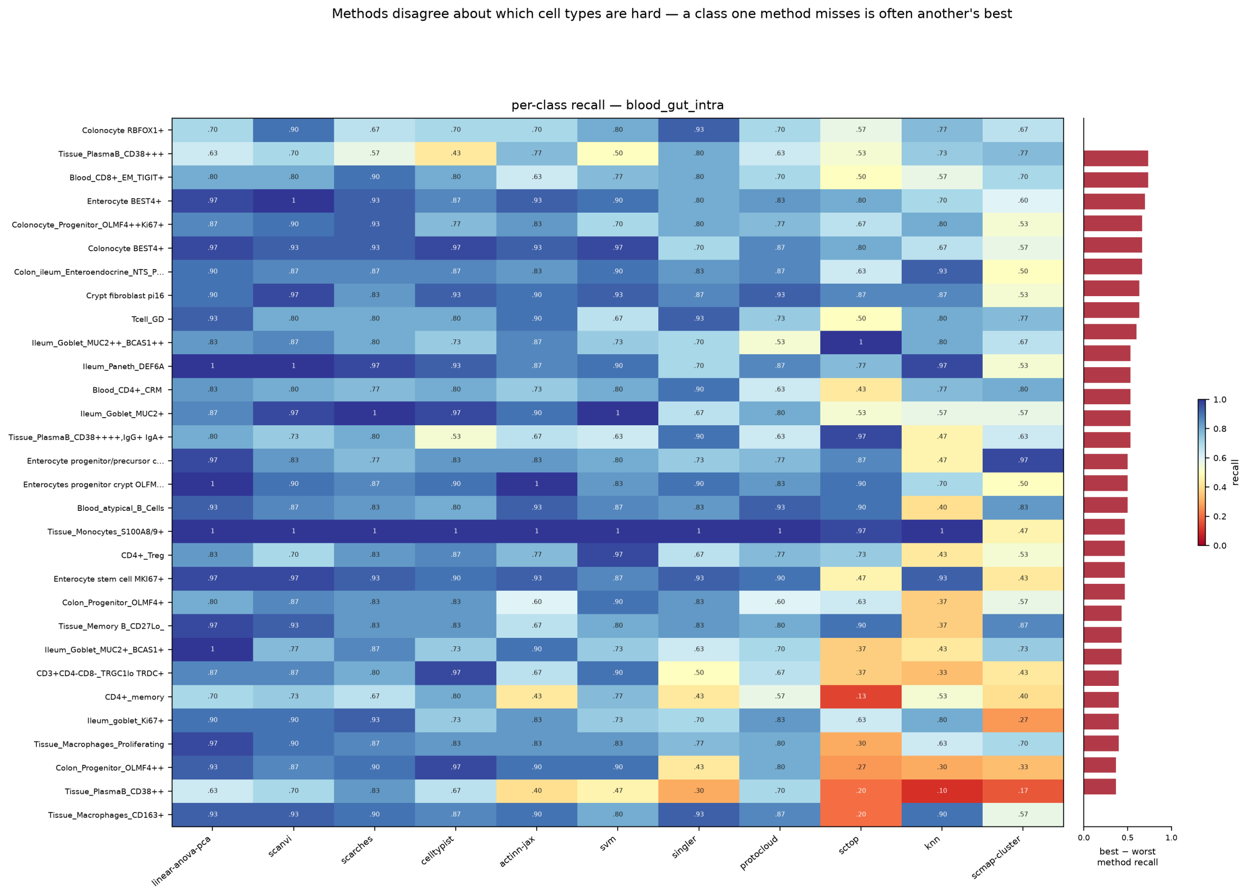

Figure S2. Per-class recall for eleven methods on the 86-type blood+gut split, ordered by how much the methods disagree (best minus worst recall, right-hand strip). The best and worst method differ by a median of 0.30 recall per class and up to 0.73; mean pairwise Spearman between methods' per-class recall is 0.58. Class size is not the explanation, and on this split it cannot be: capping per label leaves **84 of the 86 classes holding exactly 30 test cells**, with the remaining two at 29 and 14. There is no size variation left for rarity to act through, so the disagreement is about which types a method finds hard, not how many examples it saw.

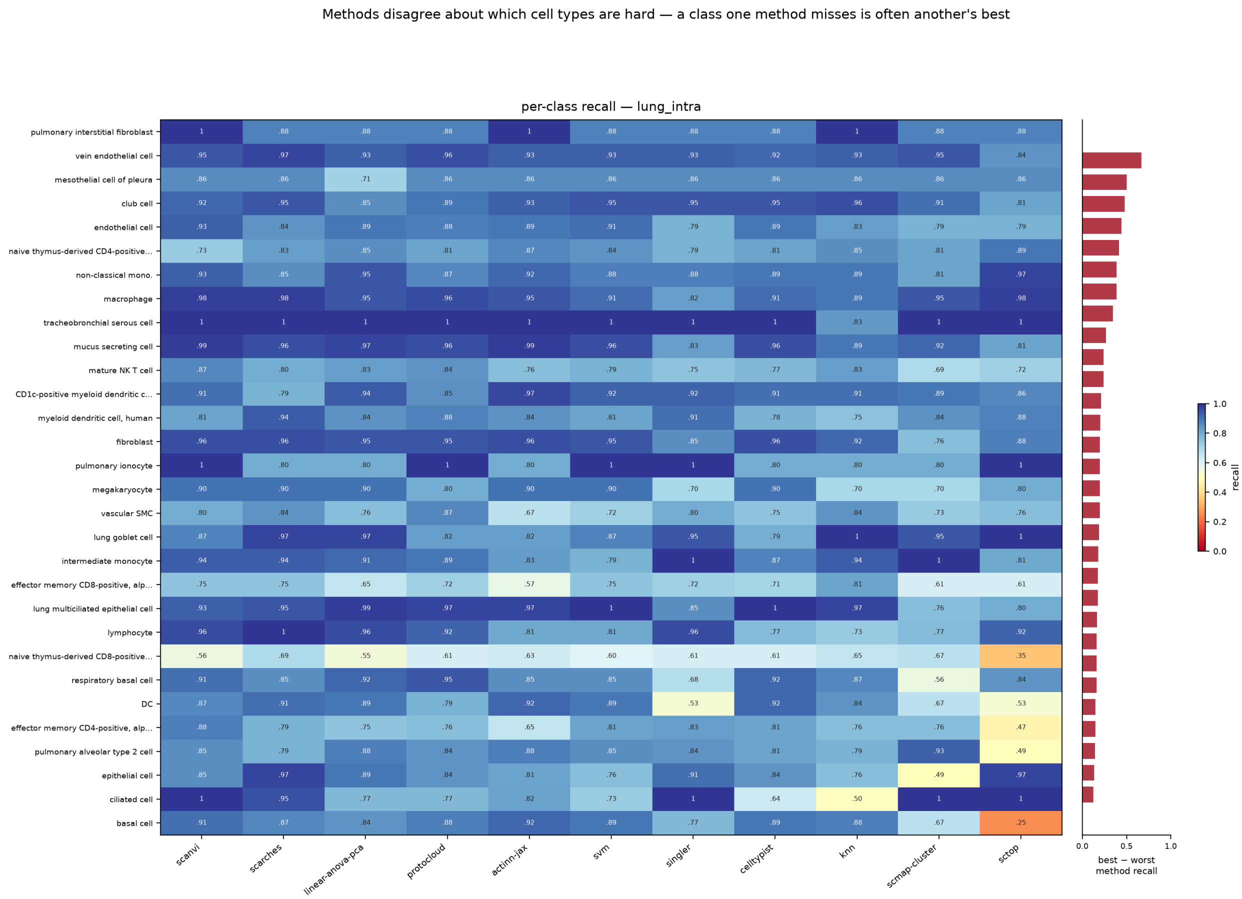

Figure S3. Per-class recall on the 46-type lung split, drawn as in Figure S2. Methods agree more here than on blood+gut: mean pairwise Spearman 0.65, median best-minus-worst 0.16.

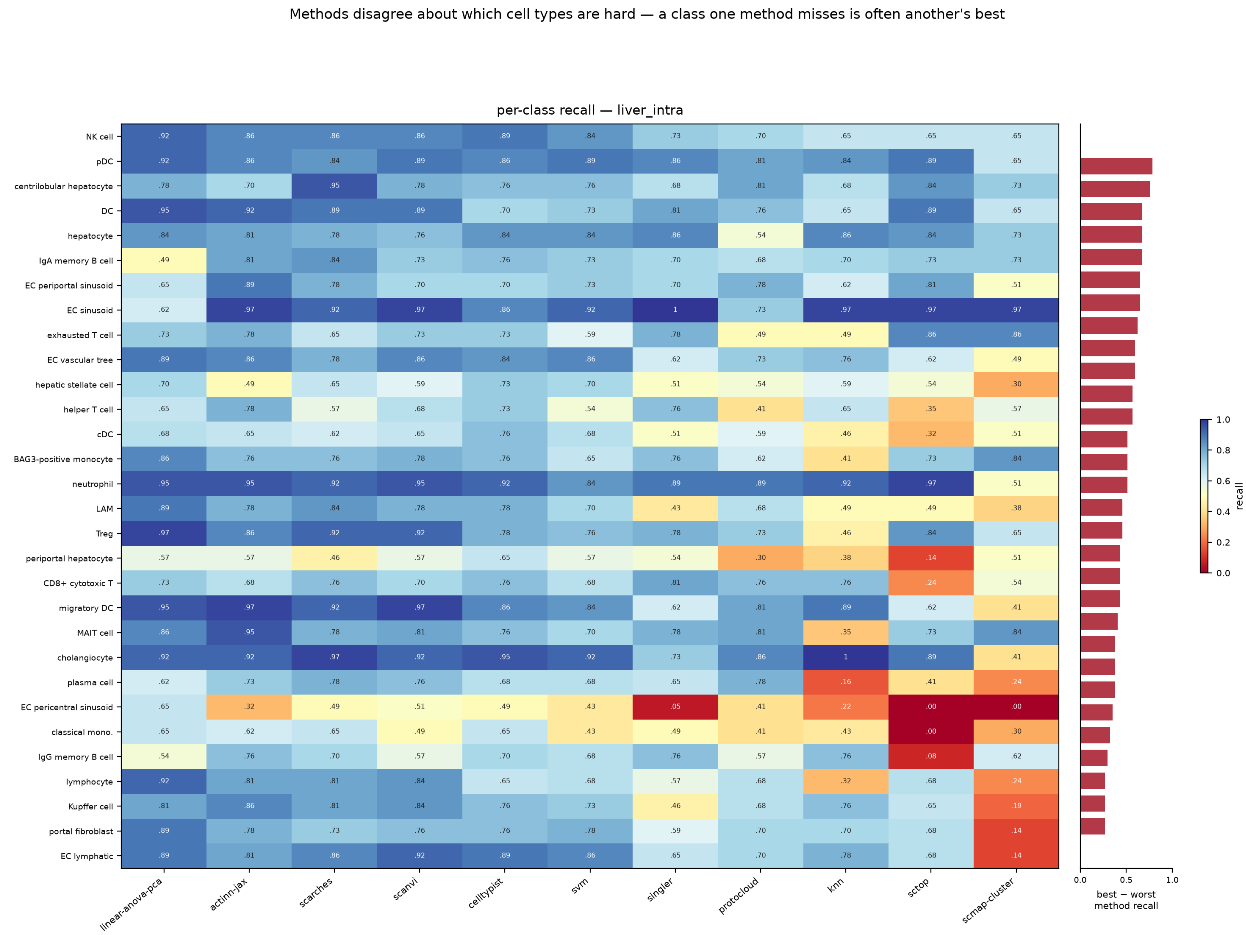

Figure S4. Per-class recall on the 36-type liver split, drawn as in Figure S2. Mean pairwise Spearman is 0.66, but the median best-minus-worst gap is 0.43 — the widest disagreement of the three splits, and a reminder that a high rank correlation between methods still leaves large per-class differences.

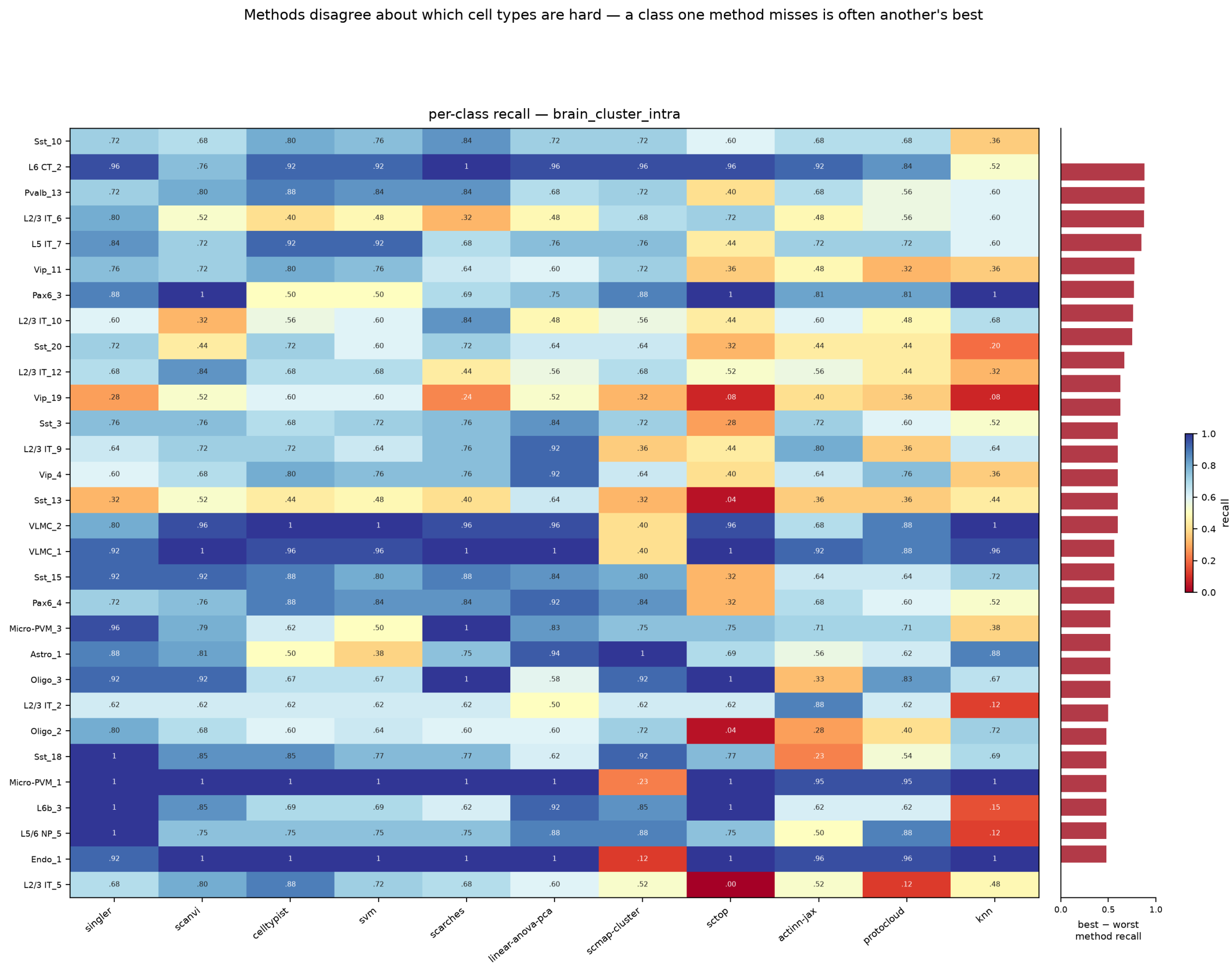

Figure S5. Per-class recall on the 151-type brain cluster split, drawn as in Figure S2. This is the split where the panel's ranking inverts (§3.1): the two correlation methods, weakest almost everywhere else, sit among the leaders, and actinn-jax is tenth of eleven. Mean pairwise Spearman is 0.61 and the median best-minus-worst gap is 0.32, rising to 0.88. The disagreement is concentrated in the graded families — `L2/3 IT`, `Sst`, `Vip` — where neighbouring clusters differ by degree rather than by marker, and not in the rare classes: capping at 100 cells per cluster leaves 130 of the 151 classes tied at 25 test cells.

Supplementary tables

method	source	tier	engine	rejection	runtime
actinn-jax	[Ma & Pellegrini 2020]	classical	JAX MLP	confidence threshold	JAX
SVM	[Pedregosa 2011]	classical	linear SVM (SGD)	—	scikit-learn
kNN	[Pedregosa 2011]	classical	k-nearest neighbors	—	scikit-learn

method	source	tier	engine	rejection	runtime
CellTypist	[Domínguez Conde 2022]	linear	L2 logistic regression	prob threshold	scikit-learn
linear-anova-pca	[Pedregosa 2011] †	linear	normalize → ANOVA → PCA(220) → logreg	prob	scikit-learn
scTOP	[Yampolskaya 2023]	parameter-free	rank z-score class-average projection	—	NumPy
SingleR	[Aran 2019]	correlation	Spearman + fine-tuning	—	R
scmap-cluster	[Kiselev 2018]	correlation	centroid cosine	yes (unassigned)	R
scANVI	[Xu 2021]	deep	scVI semi-supervised VAE	prob	scVI
scArches	[Lotfollahi 2022]	deep	scANVI reference surgery	prob	scVI
ProtoCloud	[Guo & Ding 2026]	deep	prototype VAE + LRP attribution	ambiguity flag	PyTorch
scPRINT	[Kalfon 2025]	foundation	pretrained transformer, zero-shot	—	PyTorch
Pan-human Azimuth	[Sarkar et al. 2026]	pretrained	8-level hierarchical NN, fixed 382-leaf typology	trained `Unassigned`	TensorFlow

Table S1. The benchmarked methods: model family (*tier*), the engine each one runs, whether it can decline to call a cell (*rejection*), and the runtime stack it was run on. Bold marks the methods added to this panel here.

† `linear-anova-pca` has no method paper: it is a baseline assembled here from scikit-learn components, tuned deliberately to be the strongest simple competitor we could build.

dataset	source	split	tissue	#types	genes	notes
lung_intra	[Travaglini 2020]	within-dataset	lung (Krasnow)	46	Ensembl	300 cells/type ref
lung_cross	[Sikkema 2023] → [Travaglini 2020]	cross-dataset	lung (HLCA → Krasnow)	46	Ensembl	different lab/protocol
liver_intra	[Edgar 2026]	within-dataset	liver (HLiCA)	36	Ensembl	150 cells/type
liver_cross	[Edgar 2026]	cross-**study**	liver (HLiCA)	34	Ensembl	train 6 studies → test withheld study

dataset	source	split	tissue	#types	genes	notes
blood_gut_intra	[Alegbe 2026]	within-dataset	blood + gut (IBDverse)	86	Ensembl	high cardinality; no CL ids
pbmc	[10x Genomics 2016]	within-dataset	PBMC (pbmc3k)	8	symbols	small-n; scPRINT skips (symbols)
brain_intra	[Jorstad 2023]	within-dataset	brain, MTG (Allen)	24	Ensembl	single nuclei; Allen subclasses
brain_cluster	[Jorstad 2023]	within-dataset	brain, MTG (Allen)	151	Ensembl	same nuclei, CCN cell sets

Table S2. The eight benchmark splits over seven datasets. *split* separates within-dataset holdouts from cross-dataset and cross-study transfer; *genes* records whether the matrix is keyed by Ensembl ids or gene symbols. Brain is labelled with Allen's `Subclass` rather than with Cell Ontology terms, which collapse its 24 subclasses to 18 (§2.3), so it carries no ontology score.

dataset	actinn-jax	best method (acc)
lung_intra (46 types)	0.894	ProtoCloud 0.932
lung_cross (cross-dataset)†	0.358 exact / 0.752 ontology	ProtoCloud 0.374 / **0.791 ontology**
liver_intra (36 types)	0.802	linear-anova-pca 0.804
liver_cross (cross-study)	**0.686 exact / 0.731 ontology**	actinn-jax
blood_gut (86 types)	0.860	linear-anova-pca 0.902
pbmc (8 types)	0.913	scArches 0.940
brain, subclass (24 types)	0.986	ProtoCloud 0.991
brain, cluster (151 types)	0.741	SingleR 0.845

Table S3. Per-dataset accuracy: actinn-jax against whichever method leads that dataset. † on lung_cross the exact-match score is a vocabulary artifact, so ontology concordance is the meaningful column.

Cost and scaling

These studies establish the cost claims the main text summarizes in one paragraph each. They are here rather than in the main text because the workflow argument depends on inference being cheap, not on the shape of the curve that makes it cheap.

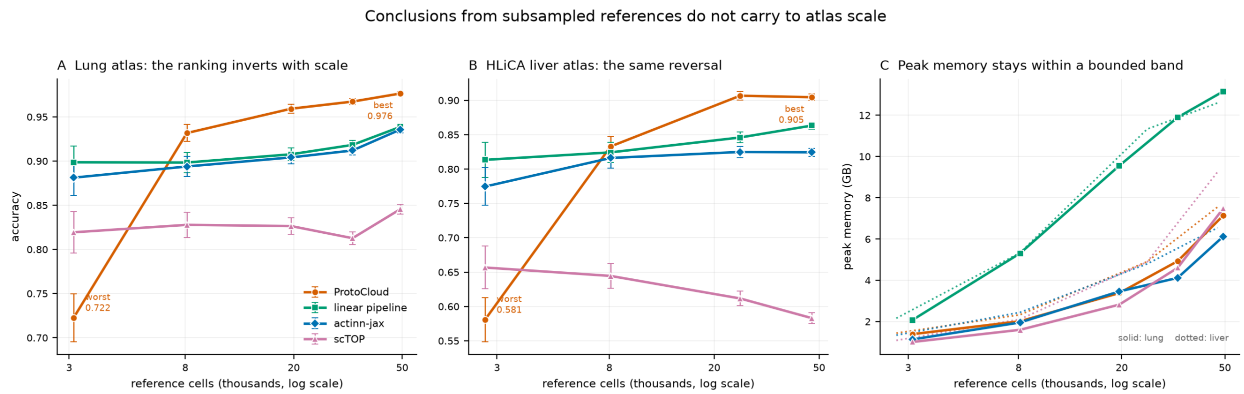

Figure S6. Accuracy and peak memory against reference size, four methods carried to full atlas scale on two datasets. *A:* lung, 3k → 49k reference cells. *B:* the HLiCA liver atlas, 2.7k → 47k, an independent replication of the same reversal — a prototype VAE moves from worst to best as the reference grows. *C:* peak memory over both sweeps. Error bars on *A* and *B* are 95% binomial intervals on the query, which grows with the reference (1,035 to 16,398 cells on lung), so they narrow from left to right; each point is a single run, so they cover sampling of the query and not run-to-run variation.

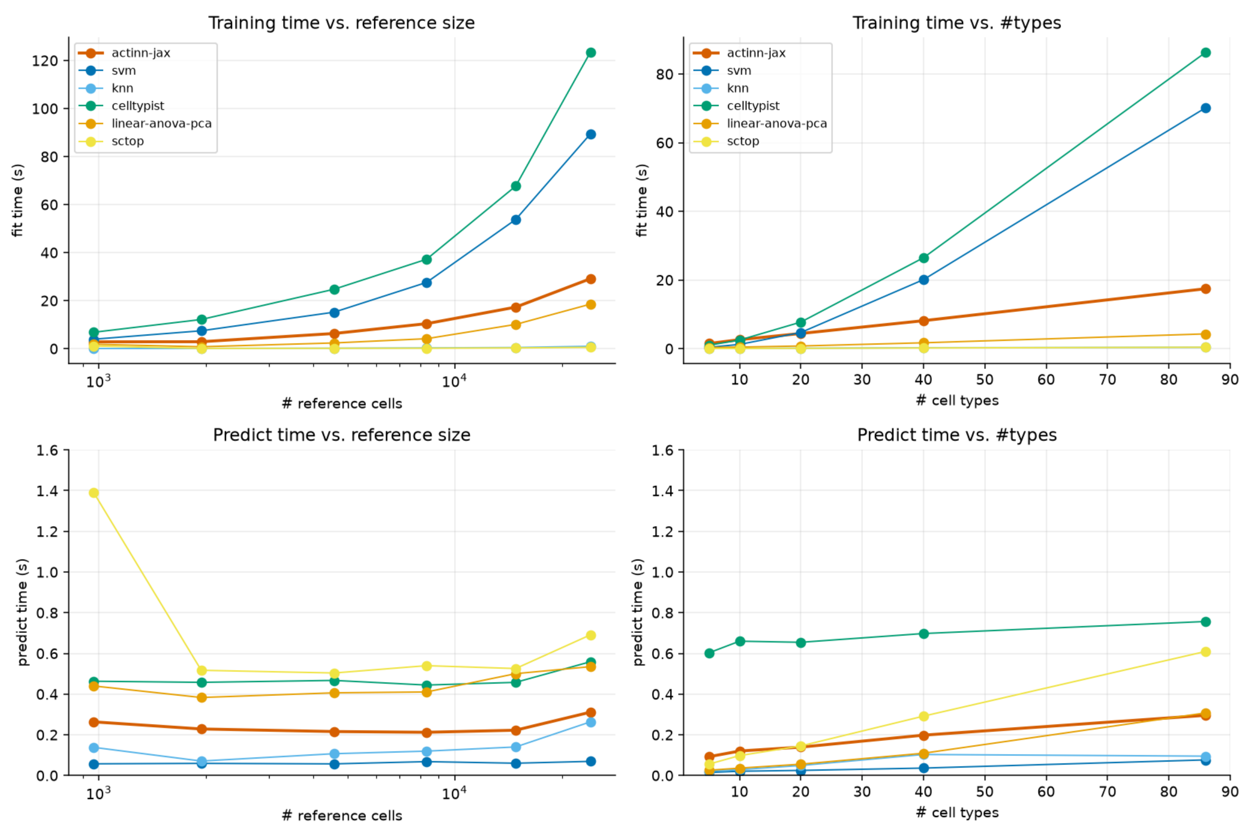

Figure S7. Fit and predict time against reference size and label cardinality, for the six CPU methods. Fit grows on both axes for every trained method, steeply for CellTypist and the SVM. Predict is flat in reference size for all six, so that flatness is a property of cached inference generally rather than of any one method; cardinality is the axis that separates them, where actinn-jax rises 3.2× from 5 to 86 types against the tuned linear pipeline's 11.9×. SVM and kNN predict faster than either throughout. scTOP's smallest-reference point carries one-time import cost and is not a scaling effect.

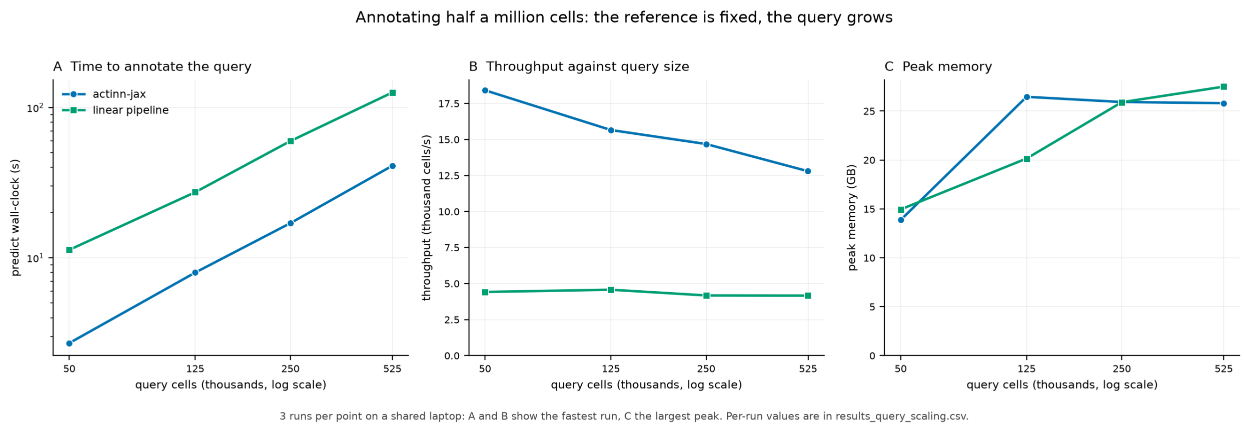

Figure S8. Cost against query size — 50,000 to 524,699 cells — with the reference fixed at 17,753 cells; every query in the main accuracy panel is smaller than the leftmost point here. *A:* wall-clock to annotate the query. *B:* throughput, which declines 31% for actinn-jax and holds flat for the linear pipeline, narrowing the advantage from 4.2× to 3.1×. *C:* peak memory, which does not distinguish them — on this axis it measures holding the query, not running the method. Three runs per point on a shared laptop: *A* and *B* report the fastest run, since contention can only add time, and *C* the largest peak, since resident-set size understates a run the OS has partly evicted.

Abstention

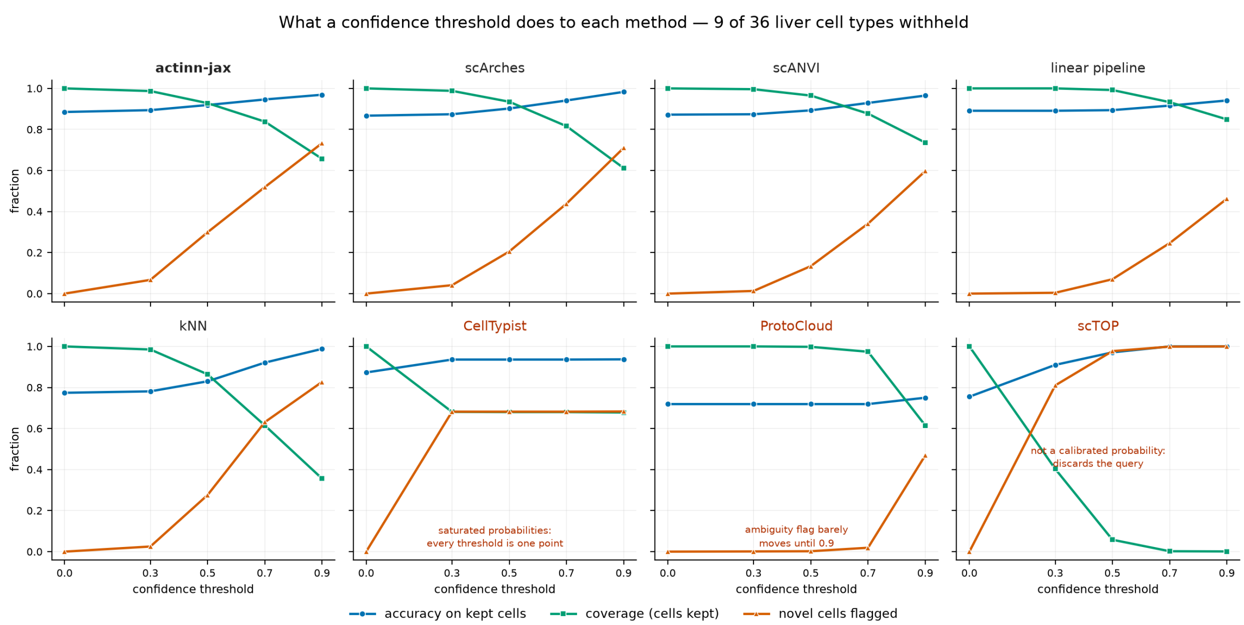

Figure S9. What a confidence threshold does to each of the eight methods that return a per-cell confidence, with 9 of 36 cell types held out of the liver reference so 1,350 query cells are out-of-distribution. Every quantity is a fraction, so all three share one axis. The five that work share a shape — accuracy and novelty rising as coverage falls — and the three that do not each fail visibly and differently.

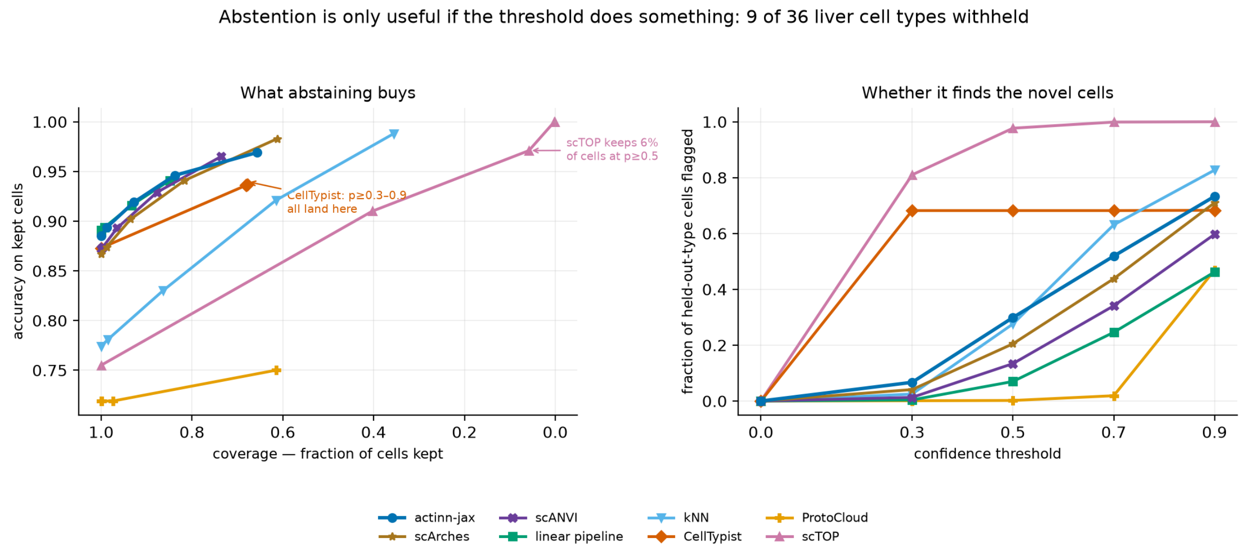

Figure S10. The same sweep read as a trade-off, which is what compares methods to each other rather than to themselves. *Left:* accuracy on kept cells against the fraction kept — the five usable methods lie along one band, which is the basis for calling their abstain quality tied. *Right:* the share of held-out-type cells flagged as the threshold rises.

Per-split accuracy and cost

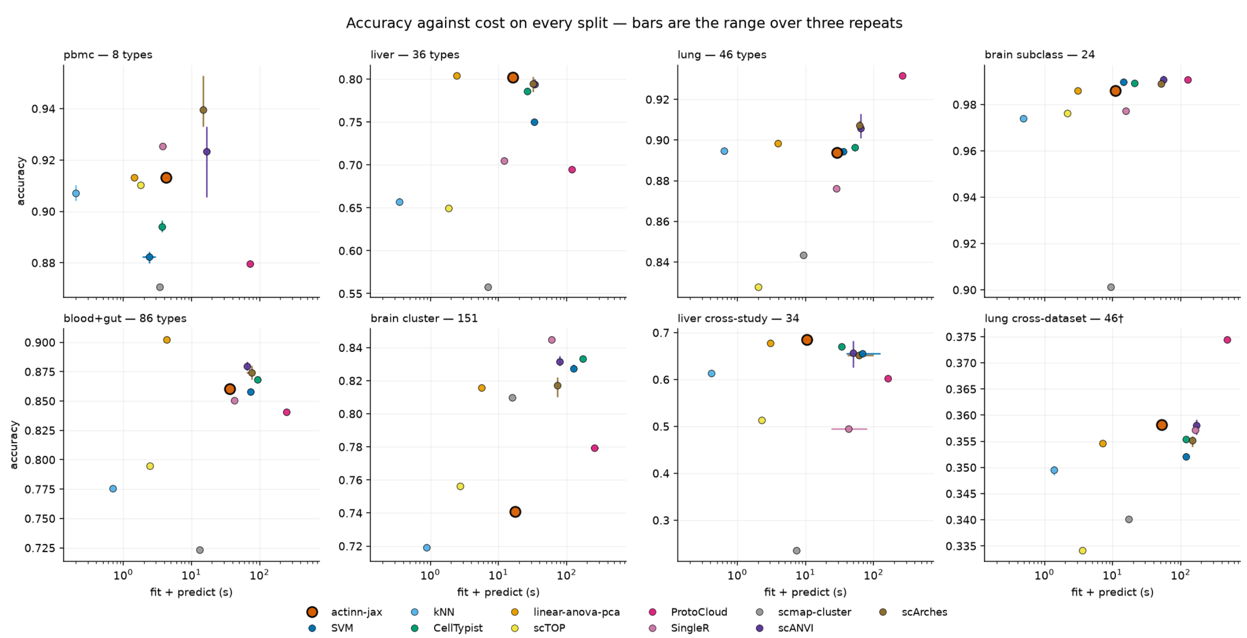

Figure S11. The data behind Figure 1 without the gap-to-leader normalization: accuracy against fit + predict time on each of the eight splits, error bars the range over three repeats on both axes. Axis limits are per panel, so vertical position compares methods within a split and not across them — the splits differ by more than half an accuracy point end to end (pbmc 0.87–0.94, cross-dataset lung 0.33–0.38†), which is why Figure 1 normalizes. Most error bars are invisible because most methods are deterministic given a fixed split; the ones that show are scANVI and scArches, whose repeats differ by up to 0.056 and 0.020. † cross-dataset lung exact-match accuracy is a label-vocabulary artefact (§3.1).